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LESSON PLAN	**COURSE: MHF4U**
Date: Lesson #12	**Unit of Study:** - **Chapter #6 Exponential and Logarithmic Functions** - **Chapter#7 Tools and Strategies for Solving Exponential and Logarithmic Equations**
Materials / Resources: - **Advanced Function 12 Study Guide** - **Chapter # 6 & 7 Formula, Aid and Homework Sheet(s)**	**Lesson Focus** - **Transformation of logarithmic functions** - **Power Laws of Logarithm** - **Equivalent forms of Exponential Equations**

Prior Knowledge:

- Graph transformations
- Logarithm techniques
- Basic procedures to solving exponential functions

Overall Expectations:

1. Demonstrate an understanding of the relationship between exponential expressions and logarithmic expressions, evaluate logarithms, and apply the laws of logarithms to simplify numeric expressions
2. Identify and describe some key features of the graphs of logarithmic functions, make connections among the numeric, graphical, and algebraic representations of logarithmic functions, and solve related problems graphically
3. Solve exponential and simple logarithmic equations in one variable algebraically, including those in problems arising from real-world applications

Specific Expectations:

2.3 Determine, through investigation using technology, the roles of the parameters d and c in functions of the form $y = (x - d) + c$ and the roles of the parameters a and k in functions of the form $y = a(kx)$, and describe these roles in terms of transformations on the graph of $f(x) = x$ (i.e., vertical and horizontal translations, reflections in the axes; vertical and horizontal stretches and compression to and from the x- and y-axes)

3.1 Recognize equivalent algebraic expressions involving logarithms and exponents, and simplify expressions of these types

3.2 Solve exponential equations in one variable by determining a common base (e.g., solve $4^x = 8^{x+3}$ by expressing each side as a power of 2) and by using logarithms (e.g., solve $4^x = 8^{x+3}$ by taking the logarithm base 2 of both sides), recognizing that logarithms base 10 are commonly used (e.g., solving $3^x = 7$ by taking the logarithm base 10 of both sides)

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Learning Goals:

- Students will be able to apply transformation to logarithmic functions
- Students will be able to apply the power law of logarithms
- Students will know how to solve for an unknown exponent of an exponential equations by applying the power law of logarithms
- Students will be able to apply the change of base formula to evaluate a logarithm having any base
- Students will be able to apply the change of base formula to graph a logarithmic function having any base
- Students will be able to express a power using different bases

Success Criteria:

1. By the end of this lesson I will be able to apply transformation to logarithmic functions
2. By the end of this lesson I will be able to apply the power law of logarithms
3. By the end of this lesson I will be able to solve for an unknown exponent of an exponential equation by applying the power law of logarithms
4. By the end of this lesson I will be able to apply the change of base formula to evaluate a logarithm having any base
5. By the end of this lesson I will be able to apply the change of base formula to graph a logarithmic function having any base
6. By the end of this lesson I will be able to write a power using different bases

Assessment Strategies:

<u>Formative</u>: homework for chapter 6.3, 6.4 & 7.1

<u>Summative</u>: quiz #5, test #4 & assignment #1

Time: **Part 1**	**Introduction:** Take up / check homework from lesson #11 and check for student's understanding. If needed, go over specific questions.
Time: **Part 2**	**Lesson:** <u>Chapter 6.3 Transformation of Logarithmic Functions</u> Based on previous functions and their transformations, evaluate the similarities and differences to transform a logarithmic function.Go over each individual parameter to explain what each stands for (e.g., for function $y = a \log log\,[k(x - d)]\ +\ c$, *a* = vertical stretch / compression, *k* = horizontal stretch / compression, *d* = horizontal translation, *c* = vertical translation), and what it means for *a*&*k* to be less or greater than 1. Ensure the steps to applying transformation. First, make sure the function is in the form $y = a \log log\,[k(x - d)]\ +\ c$. Second, apply any horizontal / vertical stretches or compressions. Third, apply any reflections. Fourth, apply any horizontal or vertical translations.

	Chapter 6.4 Power Laws of Logarithms Use the formula sheet to introduce the power laws of logarithms and apply proofs to justify these laws. Apply the laws to evaluate and solve logarithmic equations. Identify restrictions for the laws (e.g. $x^{n} = nx$ for $b > 0,\ b \neq 1,\ x > 0,\ \& n \in Rand$ $m = \frac{\log log\ m}{\log log\ b},\ b > 0,\ b \neq 1,\ m > 0$). Chapter 7.1 Equivalent Forms of Exponential Equations Demonstrate by using simple examples that exponential functions and expressions can be expressed in different ways (e.g. $9 = 3^{2}$).Identify patterns within equations, (e.g. if bases can be expressed by a same smaller base) and ways of manipulating exponents and roots to benefit solving an expression. Explain the process of solving for 'x' with different functions on each side of the equation and therefore, how it is solving for an intersecting point with the 2 functions.
Time: **Part 3**	**Wrap-up:** Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.

Notes, reminders & homework:

Reminder:

- Quiz #5 (to be written before / after lesson #13)
- Students should be receiving their assignment #1 after current class. Select one of two. Students have one week from current date to finish the assignment.

Homework:

Chapter 6.3: # 1 – 8, 12, 16
Chapter 6.4: # 1 – 13, 15
Chapter 7.1: # 1 – 5, 7, 10, 14, 15

Lesson Reflection:

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